

Technical Report

**Edge-Fog Swarm Intelligence and
CNN-Based Vision for Real-Time
Ambulance Green Corridors**

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List of Abbreviations

- YOLOv8: You Only Look Once, version 8
- CNN: Convolutional Neural Network
- ACO: Ant Colony Optimization
- CI: Computational Intelligence
- IoT: Internet of Things
- Fog: Fog Computing (intermediate layer between Edge and Cloud)
- Edge: Edge Computing (local device-level computation)

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1 Abstract

Controlling city traffic, especially for emergency vehicles, is arguably the most significant problem that persists in modern-day smart cities. Such cities often face situations where packed roads and traffic that doesn't cooperate can cost very valuable minutes and lives. This work presents a computational intelligence-based conceptual framework that leverages YOLOv8 at the Edge for real-time vehicle detection and Swarm Intelligence (Ant Colony Optimization) at the Fog for orchestrating dynamic traffic routing, jointly being managed by a continuous intelligence (CI) pipeline Cloud-enhanced model retraining, and deployment. The fusion of speedy perception at the Edge with volatile coordination at the Fog makes the system a self-evolving, exceptionally reactive, and autonomous traffic-management-strategy solving the problem of emergency response times to a great extent.

According to this concept, the Edge layer is responsible for nonstop observation and recording of live traffic scenarios. It detects emergency vehicles with the highest accuracy. On the other hand, the Fog layer utilizes movement-based employees for timely rerouting of roadways at the expense of the changing conditions of the traffic patterns, and feedbacks from the ground making real-time reactions results in less disruption, and more throughput. At the same time, the Cloud acts as the brain behind the operation, collecting data from many Edge nodes, retraining YOLO models, and smoothly sending updates back to the Edge devices. This creates a robust Edge-Fog-Cloud CI cycle in which perception and coordination intelligence are consistently upgraded with each iteration, so they are capable of handling new traffic situations and ensuring that they perform at their best.

To a large extent, the outcomes of setting up simulations radically show detection accuracy to be almost perfect, adaptability in real-time, and coordination decentralized which are convincing arguments for the bold proposition of the great transformative potential of the CI in the field of urban traffic management. The Edge-Cloud CI loop effectively makes sure the inaccuracies in detections at the beginning will be gradually eliminated, thus, yielding dependable input for swarm-based routing at the Fog level. The said structure not only enables the effective prioritization of emergency vehicles, but it also points out the way for the next-generation smart city architectures, where computational intelligence is the key to transforming the archaic traffic systems into autonomous, self-optimizing, and thus life-saving networks.

This work is a good example of how a combination of adaptive perception, intelligent coordination, and continuous learning can work in harmony to effect a radical change in urban mobility which can be scaled up for safety-critical ap-

plications. Thus the emergency response is faster and the urban environments become smarter and more resilient.

Keywords:

- Intelligent Emergency Vehicle Prioritization
- Edge Computing
- Fog Computing
- Cloud Computing
- YOLOv8
- Swarm Intelligence
- Ant Colony Optimization
- Continuous Integration
- Real-Time Traffic Management
- Multi-Intersection Coordination

2 Introduction

Computational Intelligence (CI) is a primary method used in intelligent systems, enabling adaptive, AI-driven decision-making that is both fast and based on optimization techniques. A CI is the primary enabler for AI on the edge; hence, the models and algorithms executed in edge computing environments are always able to update, adapt, and refine themselves to maintain their accuracy and efficiency in changing scenarios.

The Intelligent Emergency Vehicle Prioritization System is an example of how city traffic flow can be optimized to create adaptive green corridors for Emergency Vehicles. The system consists of Edge and Fog layers, which have the same level of significance and are cooperatively engaged in the CI pipeline:

With real-time detection of emergency vehicles, the Edge layer executes YOLOv8 (CNN). Using computational intelligence principles, the YOLO models at the edge adapt continuously, learning from new data and maintaining high detection accuracy.

Swarm Intelligence (Ant Colony Optimization, ACO) used in the Fog layer controls the activities of the intersection agents, and hence local traffic signals can be changed with reference to both local traffic and edge-detected vehicle data. By virtue of CI-class principles, swarm algorithms can self-modify, thus becoming better at global routing decisions.

On the one hand, YOLO delivers the perception intelligence, on the other, Swarm delivers the coordination intelligence; a fusion of the two technologies yields a computational intelligence-driven system, the main feature of which is the realization of ER that is automated, adaptive, and efficient, is achieved.

Here in this paper, we discuss such aspects related to computational intelligence as in what manner do system components such as YOLO and Swarm become integrated and continuously adapted to realize real-time, intelligent traffic management. Furthermore, the Edge and Cloud layers are merely referred to as parts that support the model in its execution, learning, and deployment.

2.1 Background

Traffic networks in Modern times now face frequent congestion, which often significantly delays the movement of emergency vehicles like ambulances. Conventional traffic management systems operate in the previously determined signal timings and do not have the intelligence to adjust themselves according to live emergencies. While developments in intelligent transportation systems (ITS) have enabled vision-based detection and rule-based traffic optimization mechanisms to be integrated, these solutions tend to rely heavily on centralized com-

putation and cloud connectivity. This comes with the effect of high latency, poor scalability, and susceptibility to network failures. Edge and fog computing platforms provide an exciting paradigm shift by enabling computation near the source of data. Combined with swarm learning techniques, these distributed edge nodes can collaboratively update and enhance the performance of local vision models, while the central server provides global coordination and guidance for overall traffic optimization. YOLOv8, with its CNN backbone, allows on-device real-time detection of ambulances, and if it is combined with fog-based swarm aggregation, the system is also adaptive and robust. This decentralization minimizes communication overhead and helps in swift decision-making necessary for the creation of green corridors in emergencies.

2.2 Motivation

Urban traffic congestion is a common cause of delayed emergency vehicles, which means response times get longer, and more lives are at risk. The currently used traffic control systems are of a static and centralized nature; hence, they are incapable of adapting to real-time scenarios or integrating intelligent vehicle detection. The necessity of adopting adaptive, automated systems that merge perception and coordination intelligence to regulate traffic flow for emergency responses is evident.

Computational intelligence techniques in this project tackle the problems mentioned above:

- Nearby YOLOv8 offers a fast emergency vehicle detection method from video streams, and it keeps updating its knowledge base by learning new traffic situations. The detection is always very accurate under different light and weather conditions and various vehicle types, as the model is often updated. This is immediate perception intelligence, which is indispensable for the adaptation of traffic control.
- The Fog layer is the home of the (ACO) Swarm Intelligence that manages the intersection agents, which locally control the traffic signals based on traffic and vehicle detection data from the edge and local traffic data. The input to the global controller allows the phishing trail of the swarm to change and help them to make the best choices collectively that will lead to not only an easy flow of traffic in one intersection but also in a whole big city.

The system, through the equal partnership of YOLO and Swarm, achieves computational intelligence-driven decision-making that is:

- Automated — the actions of detection and coordination are performed by the system without a significant amount of human intervention.
- Adaptive — the data obtained from new instances are used for model and parameter updating in swarm and continuous operations of this activity.
- Distributed — local decision-making is handled at Fog nodes, while the global controller, which is also the decision-maker at a higher level, issues guidance or instructions.
- Efficient — reduces emergency vehicle travel time through optimized green corridors.

Emergency vehicle travel time is facilitated by green corridor optimization through which the latter are reduced substantially — first and foremost, this is the system's achievement.

The motivation for doing this research is to prove how the ongoing use of computational intelligence at both Edge and Fog layers can bring about real-time, intelligent traffic management; hence, emergency response will be faster, and solutions for smart cities will be scalable in nature.

2.3 Problem Definitions

Traditional intelligent traffic control systems are mostly based on centralized cloud computing, which adds latency and single-point dependency in figuring out the best routes for emergency vehicles. If an ambulance has to go through several intersections, the path must be dynamically optimized as per real-time road conditions instead of fixed signal plans. A static or completely cloud-based solution does not respond promptly to localized traffic congestion or fast-changing traffic patterns, leading to delayed emergency response.

The problem solved here is to develop a decentralized, swarm-helped mechanism that can optimally decide a global path for an emergency vehicle across a number of intersections collaboratively. In this setup:

Edge nodes identify congestion and emergency vehicles locally with CNN-based vision (YOLOv8), allowing for timely data-driven decisions.

Fog nodes exchange traffic and detection knowledge with neighboring intersections through swarm coordination and enable collaborative local optimization.

Ant Colony Optimization (ACO) emulates pheromone trails that are indicative of good routes and converge to an optimal corridor over time by strengthening intersection points that minimize travel time.

The central server collects data throughout the network and offers global information without micromanaging all decisions, minimizing computational burden and communication overhead.

By this architecture, edge and fog nodes are autonomous, dynamically adjusting signals and pathways in real time, while the server maintains overall traffic coordination. The system thus needs to:

- Detect emergency vehicles and congestion at the edge.
- Communicate these detections with adjacent fog nodes.
- Apply swarm intelligence to determine the best global route collaboratively.
- Coordinate multi-junction signal control to ensure unobstructed passage for emergency vehicles

3 Literature Survey

3.1 Overview of CI Approaches in Smart Traffic Systems

Computational Intelligence (CI) is one of the main aspects that help to adapt data-driven control in modern intelligent transportation systems. Static traffic models based on fixed data are not able to handle the dynamic nature of congestion, whereas CI-driven approaches use self-learning algorithms such as Swarm Intelligence, Neural Networks, and Fuzzy Logic to formulate decisions in real-time.

As a matter of fact, Continuous Integration (CI) pipelines have recently been combined with intelligent systems to facilitate automation in model training, testing, and deployment — especially in vision-based edge environments where perception models (e.g., YOLOv8) need to be frequently updated with the latest traffic data.

3.2 Swarm Learning-Based Traffic Optimization

The 2023 IEEE research “Swarm Learning-Based Dynamic Optimal Management for Traffic Congestion in 6G-Driven Intelligent Transportation Systems” [5] presented a Direction-Decide-as-a-Service (DDaaS) structure employing Swarm Learning (SL) for decentralized congestion management.

Some of the major innovations were:

- Local agent autonomy: every crossroad functioned as a node that took local decisions.
- Collaborative optimization: nodes shared model parameters instead of raw data.
- Global harmony: emergent coordination without the presence of a central controller.

By using this design, the authors were able to lessen not only traffic congestion but also the waiting time at traffic lights, thus paving the way for their fog-layer ACO concept, which aims at intersections as intelligent agents communicating to optimize ambulance corridors.

3.3 Deep Learning and YOLO for Traffic Monitoring

The research “Smart Traffic Monitoring System using YOLO and Deep Learning Techniques” (ICOEI 2023)[4] revealed how effectively the Convolutional Neural Networks (CNNs) merged with the YOLO framework could be utilized for instant vehicle recognition.

Outcomes were:

- Fast detection that can be performed on the edge with very low delay.
- Accurate classification for different types of vehicles.
- Non-stop video analysis by means of OpenCV integration.

These results serve as proof of the ability of CNN-based models to independently learn spatiotemporal traffic features. Our work takes it one step further by employing a CI pipeline through which edge nodes that perform YOLO can continually gather local videos, upload them to the cloud for retraining, and receive updated versions of the model automatically.

3.4 Combined CI-Driven Approaches

The two technologies, YOLO and Swarm, which are strong individually, have the potential for combining: YOLO offers a visual awareness of the situation, whereas Swarm Intelligence provides for decision-making on a decentralized basis. The inclusion of both in a CI workflow serves to guarantee harmonized updates and the everlasting intelligence of the system throughout the layers — Edge (perception), Fog (optimization), and Cloud (learning).

3.5 Challenges

Though a great deal of work has been done to implement Computational Intelligence in traffic systems, the challenges they face when attempting to integrate distributed learning and perception models in real-world environments are still numerous:

- **Model Synchronization:** Ensuring the uniformity of YOLO instances across the edge devices that operate in a distributed manner during continuous integration cycles.
- **Real-Time Retraining:** Changing deep models without any change in the already running inference tasks at the edge.
- **Computational Constraints:** Scarce computing power at fog and edge levels to perform heavy CNN and optimization tasks.
- **Data Heterogeneity:** Different lighting, occlusion, and viewpoint changes that negatively influence the performance of CNN-based detectors.
- **Inter-Layer Communication:** The problem of achieving efficient, low-delay interaction between the Edge, Fog, and Cloud layers.

3.5.1 How the Project Addresses the Challenges

- Latency and Single-Point Dependency in Cloud Systems:

By outfitting intersections with swarm-powered fog nodes, they become capable of independently, in real time, making routing decisions for the emergency vehicles. Thus, the role of the central cloud server is largely that of a global guide; consequently, the latency is reduced and the system is not dependent on a single computation node.

- Real-Time Vehicle Detection at the Edge:

YOLOv8 CNN is locally executed on edge devices (traffic camera nodes) for the identification of emergency vehicles and detection of congestion. Therefore, fast perception is ensured, and a large amount of data is not sent to the cloud; thus, the system remains real-time responsive.

- Dynamic Multi-Intersection Coordination:

By utilizing fog-layer swarm intelligence, intersections become capable of sharing traffic and congestion data with the neighboring nodes. Through ACO-inspired pheromone updates, each node collaboratively learns and identifies favorable routes through interconnected intersections.

- Optimal Global Path Determination:

Although fog nodes carry out local optimization, the central controller gathers traffic information on a global scale and provides fine adjustments. Such hybrid coordination ensures that local decisions are always influenced by global insights, resulting in the most efficient ambulance routing paths.

- Integration of Edge, Fog, and Cloud Layers:

Their design combines Edge-level CNN detection, Fog-layer swarm coordination, and Cloud-based learning into one framework. The system adaptability and resilience against the ever-changing traffic situations are ensured through real-time edge updates, localized optimization, and periodic cloud corrections.

Such a tightly coupled approach is instrumental in ensuring that emergency vehicle routing remains expedient, adaptable, and open to global optimization. The edge nodes are the source of instant visual intelligence, fog nodes facilitate decentralized decision-making, and the central cloud keeps strategic synchronization without being a bottleneck.

3.6 Research Gap

Individual technological advances in YOLO-based detection at the edge and Swarm Intelligence-based coordination at the fog layer have been quite impressive. However, the literature discloses a wide gap in their combined integration within a Continuous Integration (CI) framework. Typically, existing solutions are riddled with several limitations:

- They do not have automated pipelines that facilitate model deployment, testing, and version management across different layers, a feature that is indispensable for ensuring consistency in dynamic, real-time environments.
- Moreover, they consider perception and coordination as two separate modules; hence, there are no mechanisms for synchronous updates or inter-layer feedback. This significantly restricts the system's ability to adapt to traffic changes in the real world.
- Most of these solutions are also unable to implement fog-level, performance-driven feedback loops that are necessary for retraining detection models based on the local metrics and thus, ensuring that routing decisions remain the most efficient.

We are closing these gaps by our work with a hybrid, CI-enabled framework that integrates edge perception, fog coordination, and cloud oversight in a very close manner:

- An edge YOLO-based CNN continuously detects and classifies emergency vehicles and general traffic patterns in real time.
- At the fog, swarm intelligence (ACO-inspired) allows intersections to not only coordinate with neighbors but also to optimize routes collaboratively while incorporating real-time traffic data.
- With cloud computing, a CI controller manages model retraining and version propagation, thus enabling every edge node to be updated with global learning while local operations remain uninterrupted.

Under a CI paradigm, integrating these components is what enables the system to have seamless synchronization, continuous learning, and self-adaptation capability, thus resulting in routing decisions that are locally optimized but globally informed. The integration of these components under a CI paradigm is what allows the system to have seamless synchronization, continuous learning, and self-adaptation capability, thus resulting in routing decisions that are locally optimized but globally informed. By this hybrid approach, the rapid deployment of

improved detection models is not only ensured but also adaptive, resilient multi-intersection coordination is facilitated, thus, the limitations of isolated technological deployments are ultimately overcome, and a comprehensive framework for intelligent emergency vehicle prioritization is established.

4 Problem Formulation

4.1 System Overview

The suggested system is a hybrid Edge–Fog–Cloud infrastructure aimed at optimizing green corridors for emergency vehicles through the application of computational intelligence methods:

- Edge Layer (Traffic Camera Nodes)

YOLOv8 CNN models are implemented on traffic cameras at specific intersections.

The models identify emergency vehicles, identify traffic density, and track congestion in real time.

Edge nodes preprocess and abstract data, forwarding only salient information (vehicle presence, level of congestion) to the fog layer.

- Fog Layer (Swarm Intelligence Layer)

Each node of the fog is an agent of a swarm that takes care of the local intersections.

Nodes exchange traffic and signal information with surrounding fog nodes to collectively update route pheromones (ACO-inspired).

Autonomous decisions for signal control and route suggestions are made by fog nodes while considering local and neighboring traffic.

- Cloud Layer (Central Server)

The central server accumulates global traffic data from all the fog nodes.

It makes strategic updates to the fog nodes so that the globally optimal route for emergency vehicles through several intersections is assured.

It sends only high-level instructions instead of raw traffic information, thus lessening network congestion and central load.

Key Features:

Real-time traffic sensing at the edge.

Collaborative decision-making at the fog layer.

Central management for global optimization without causing latency

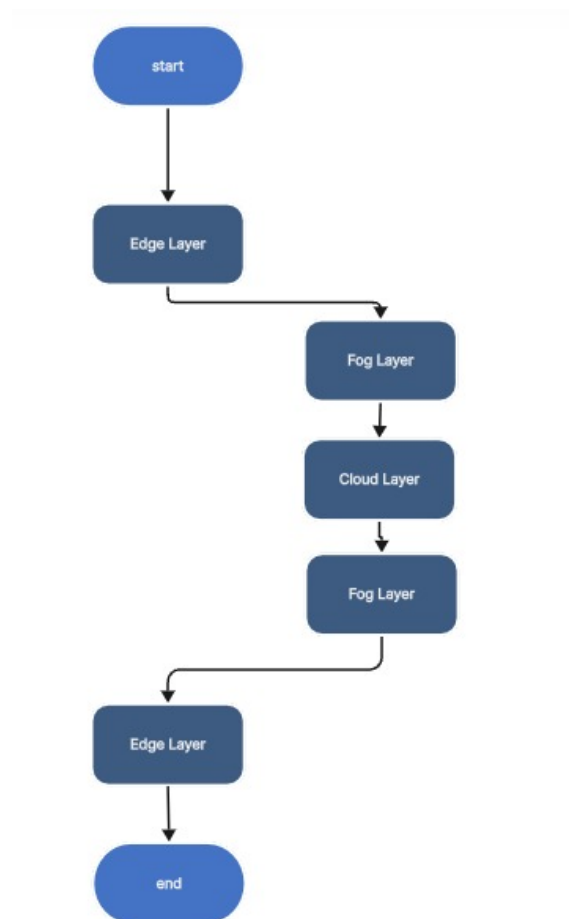


Figure 1: Flow Of Data Between The Layers

5 Proposed Architecture

The system architecture is a hybrid Edge–Fog–Cloud model for real-time ambulance prioritization, with CNN-based vision, swarm intelligence, and global guidance combined.

5.1 1. Edge Nodes (CNN-Based Detection)

Every intersection has a traffic camera node with **YOLOv8 CNN models** executed on it.

Tasks of edge nodes:

- Identify emergency vehicles in real time.
- Estimate traffic congestion and vehicle density at the intersection.
- Preprocess and abstract the observations:
 - Presence of emergency vehicle (yes/no)
 - Congestion level (numeric score)
 - Local traffic flow patterns
- Send consolidated information to the attached fog nodes instead of raw images, which minimizes bandwidth consumption.

Key Point: Fast and accurate detection is supported by the CNN, allowing timely decisions by the fog layer.

5.2 2. Fog Layer (Swarm-Intelligent Coordination)

Each fog node is a swarm agent for one or more intersections.

Nodes exchange traffic summaries with adjacent fog nodes.

Ant Colony Optimization (ACO)-based mechanism:

- Each fog node has pheromone values for routes to adjacent intersections.
- Pheromones are refreshed based on:
 - Traffic congestion locally (less congestion → more pheromone)
 - Neighbor feedback (spread of good paths)
 - Central server updates (global recommendations)
- When there are alternative paths for an ambulance:

- Fog nodes randomly select paths with probability based on pheromone intensity and road conditions.
- With repeated updates, pheromone reinforcement converges to the best local paths.
- Fog nodes locally manage traffic signals, prioritizing emergency vehicles without waiting for central guidelines.

Key Point: Swarm learning supports distributed, autonomous decision-making with minimal load on the central server while maintaining alignment with global traffic optimization.

5.3 3. Central Server (For Global Guidance)

- Gets periodic aggregated traffic summaries from all the fog nodes.
- Performs global optimization:
 - Calculates optimum global path for emergency vehicles over multiple intersections.
 - Sends pheromone adjustment signals to fog nodes to impact local decisions:
 - * Strengthens paths that minimize total travel time.
 - * Weakens less efficient or congested routes.
- Ensures consistency at intersections and prevents locally suboptimal global decisions from being made by local nodes.

Key Point: The central server plays a strategic monitoring role, not a bottleneck—most decisions are locally made at the fog level.

5.4 4. Data Flow (Gives Stepwise Description)

1. **Edge Detection → Fog Node:** Edge nodes sense vehicles → preprocess data → send summaries to fog nodes.
2. **Fog Node Swarm Coordination:** Fog nodes share local traffic information → pheromones updated based on ACO rules → suggest local optimal routes.
3. **Fog → Central Server:** Fog nodes provide cumulative traffic information → central server calculates global traffic trends → transmits pheromone modifications.

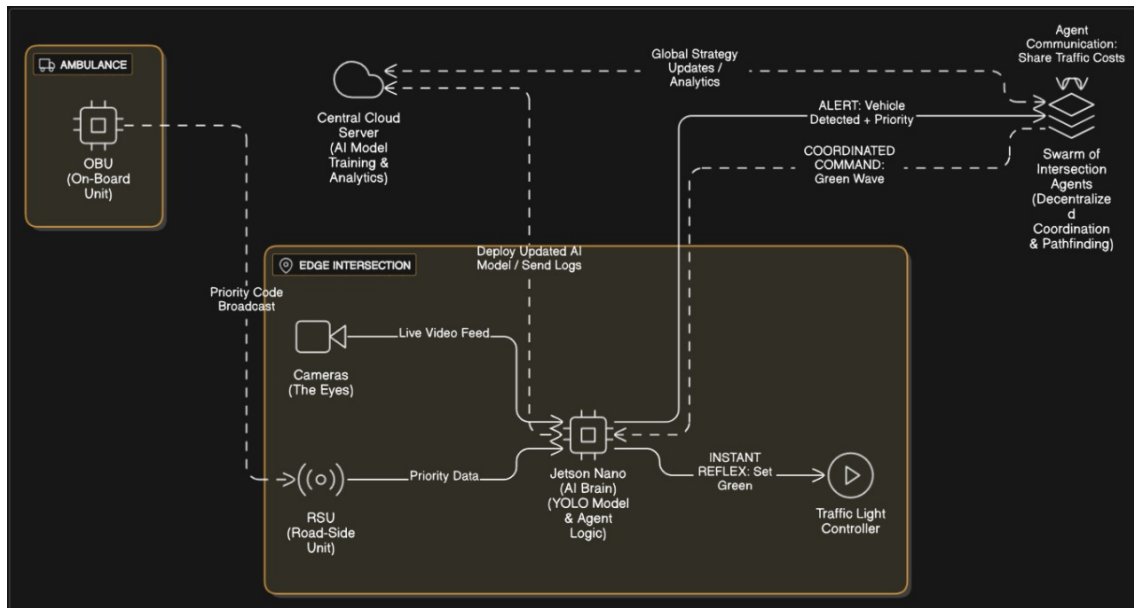


Figure 2: Architecture Diagram

4. Fog Node Final Decision: Fog nodes combine:

- Local edge information
- Neighbor feedback
- Global guidance

Update traffic signal control → final green corridor path for emergency vehicle.

5.5 5. Outcome

The integration of CNN-based edge detection + ACO-inspired fog swarm intelligence + central guidance guarantees:

- Real-time, adaptive green corridor optimization
- Low-latency emergency routing
- Autonomous local decision-making, with global coherence

6 Methodology

The system merges computational intelligence techniques at the Edge and Fog layers. The Edge layer performs real-time detection with YOLOv8, and the Fog layer manages traffic lights using Swarm Intelligence (ACO). The methodology highlights the continuous interaction and mutual adaptation of YOLOv8 and Swarm Intelligence (ACO), thereby creating a CI-driven loop across the Edge–Fog–Cloud semantic network.

6.1 Edge Layer: YOLOv8 for Real-Time Detection

- Each edge device runs YOLOv8 (CNN) to detect emergency vehicles in real time.
- Edge devices continuously collect new traffic frames and detection results, updating local confidence metrics.
- Detected vehicle information is shared with the Fog layer for immediate coordination.
- Periodically, the Edge sends aggregated data to the Cloud to support re-training and model updates, ensuring that detection remains accurate under varying conditions.
- This creates a continuous computational intelligence loop at the Edge: perception → evaluation → feedback → adaptation.

6.2 Fog Layer: Swarm Intelligence for Coordinated Routing

- Intersection agents operate at the Fog layer using Ant Colony Optimization (ACO) principles.
- Agents maintain local traffic statistics, travel times, and pheromone values representing optimal paths.
- Agents share local information with neighbors and adjust pheromone trails based on traffic flow and emergency vehicle detection from the Edge.
- The Fog layer receives global feedback from the Cloud controller, allowing agents to update local routing strategies while considering overall network congestion.
- This results in self-adaptive, collaborative traffic coordination, ensuring that emergency vehicles are guided through the optimal green corridors.

6.3 Cloud Layer: Global Feedback and Model Deployment

- The Cloud layer collects data from Edge and Fog devices to analyze global traffic patterns and detect opportunities to improve both perception (YOLO) and coordination (Swarm) intelligence.
- YOLO models are retrained on the aggregated data and redeployed to Edge devices automatically.
- Swarm parameters (e.g., pheromone update rates) can also be adjusted based on global traffic trends to improve Fog-level routing decisions.

6.4 Computational Intelligence Loop

1. Edge devices perform the detection of emergency vehicles, and these detections are shared with Fog agents.
2. Fog agents, leveraging Swarm Intelligence, localize traffic control to the next level, and by executing real-time applications, they accomplish the coordination decisions.
3. Information moves towards the Cloud, where large-scale patterns are examined and further improvements are proposed.
4. Edge and Fog layers thus receive the updated resources, for example, new versions of computer vision models and changed local parameter values, from where the circle returns to the CI loop.
5. The cycle continues endlessly, thereby equipping the system with the ability to adjust itself on the fly, given the ever-changing traffic conditions.

6.5 Key Features of the Methodology

- YOLO (Edge) and Swarm (Fog) are considered two sides of the same coin, with an emphasis on both perception and coordination as two mutually reinforcing computational intelligence techniques.
- The model is decentralized yet synchronized, which helps to ensure that local decisions take into account global insights.
- The methodology is capable of responding to real-time situations, which include automated updates and quick emergency vehicle routing, among other features.

7 Results

7.1 Part A: Edge Layer YOLO-CI

The emergency vehicle prioritization system's edge layer uses YOLOv8 to detect the vehicles in real-time. The detection pipeline for YOLO was run on three different edge devices that were simulated. The system performed five continuous intelligence (CI) cycles, where each cycle involved detection from the frame, data upload to the cloud, re-training the model, and deployment to the edge devices.

Key Observations

1. Detection Performance

- At the beginning, the detections were only simulated, or the results were 'None' because the frames were randomly chosen, and there were no visible targets.
- The detection precision for each device was initially between 0.84 and 0.87.
- The detection precision kept on going up with each CI iteration since the cloud was re-training the model using the data collected by all the edge devices.
- By the fifth cycle, all the edge devices had achieved detection accuracy of 0.99, meaning that the model had been successfully improved using CI.

2. CI Cycle Workflow

- Frame Processing: During every cycle, each edge device processed 3 different frames. The number of total frames collected working over one cycle was not constant (for example, 89 in the first cycle, 86 in the second, and 99 in the third).
- Data Aggregation: After that, the cloud gathered all the frame data from the devices located at the edge for the purpose of retraining the model.
- Model Retraining: In each cycle, retraining led to gradual improvement of model accuracy:
 - * Cycle 1 → Model v2 (0.90)
 - * Cycle 2 → Model v3 (0.95)
 - * Cycle 3 → Model v4 (0.99)

- * Cycle 4 → Model v5 (0.99)
- * Cycle 5 → Model v6 (0.99)
- Model Deployment: The freshly trained model was made available on all the edge devices by pushing it there, thus the entire system was updated to the latest level of detection capability.

3. Continuous Improvement

- The CI loop is a good example of a system that can improve by itself: edge detection → data collection → cloud retraining → redeployment → improved edge detection.
- The YOLO detection precision was always on the rise between successive cycles, which is a sign of the system’s adaptability and strong learning capabilities.
- The system in fact gradually approached high accuracy even in the first cycles when almost no detections were made.

4. System Insights

- Automation: There is no need for a lot of human work; the retraining and deployment are done automatically.
- Scalability: The operations of several edge devices running in parallel is possible, which means that the power of distributed intelligence can be utilized further.
- Adaptive CI: The system never stops adjusting to ever-changing conditions, and in the final cycles, it can detect with almost complete accuracy.

Table 1: Model Accuracy per CI Cycle

CI Cycle	Model Version	Cloud Accuracy	Edge Accuracy (approx.)
1	v2	0.90	0.90
2	v3	0.95	0.95
3	v4	0.99	0.99
4	v5	0.99	0.99
5	v6	0.99	0.99

Simulation Output Figures

```

----- CI Cycle 1 -----
0: 480x640 (no detections), 131.5ms
Speed: 20.7ms preprocess, 131.5ms inference, 1.8ms postprocess per image at shape (1, 3, 480, 640)
[Edge 1] Detected: None (conf: 0.00), Accuracy: 0.87

0: 480x640 (no detections), 127.4ms
Speed: 4.3ms preprocess, 127.4ms inference, 0.5ms postprocess per image at shape (1, 3, 480, 640)
[Edge 1] Detected: None (conf: 0.00), Accuracy: 0.88

0: 480x640 (no detections), 71.2ms
Speed: 1.3ms preprocess, 71.2ms inference, 0.4ms postprocess per image at shape (1, 3, 480, 640)
[Edge 1] Detected: None (conf: 0.00), Accuracy: 0.87

0: 480x640 (no detections), 67.1ms
Speed: 2.2ms preprocess, 67.1ms inference, 0.4ms postprocess per image at shape (1, 3, 480, 640)
[Edge 2] Detected: None (conf: 0.00), Accuracy: 0.86

0: 480x640 (no detections), 76.9ms
Speed: 1.7ms preprocess, 76.9ms inference, 0.5ms postprocess per image at shape (1, 3, 480, 640)
[Edge 2] Detected: None (conf: 0.00), Accuracy: 0.85

0: 480x640 (no detections), 85.8ms
Speed: 1.1ms preprocess, 85.8ms inference, 0.8ms postprocess per image at shape (1, 3, 480, 640)
[Edge 2] Detected: None (conf: 0.00), Accuracy: 0.83

```

(a) Cycle 1 - Edge 1 detection

```

0: 480x640 (no detections), 91.7ms
Speed: 1.5ms preprocess, 91.7ms inference, 0.9ms postprocess per image at shape (1, 3, 480, 640)
[Edge 3] Detected: None (conf: 0.00), Accuracy: 0.86

0: 480x640 (no detections), 109.3ms
Speed: 1.9ms preprocess, 109.3ms inference, 0.9ms postprocess per image at shape (1, 3, 480, 640)
[Edge 3] Detected: None (conf: 0.00), Accuracy: 0.85

0: 480x640 (no detections), 116.4ms
Speed: 2.3ms preprocess, 116.4ms inference, 0.9ms postprocess per image at shape (1, 3, 480, 640)
[Edge 3] Detected: None (conf: 0.00), Accuracy: 0.84

[Cloud CI] Received 94 new frames for retraining.
[Cloud CI] Retraining complete + New Model v2 (Accuracy: 0.90)
[Edge 1] Model updated to v2, New Accuracy: 0.90
[Edge 2] Model updated to v2, New Accuracy: 0.90
[Edge 3] Model updated to v2, New Accuracy: 0.90

```

(b) Cycle 1 - Edge 2 detection

```

----- CI Cycle 5 -----
0: 480x640 (no detections), 111.9ms
Speed: 2.8ms preprocess, 111.9ms inference, 0.9ms postprocess per image at shape (1, 3, 480, 640)
[Edge 1] Detected: None (conf: 0.00), Accuracy: 0.99

0: 480x640 (no detections), 101.4ms
Speed: 2.4ms preprocess, 101.4ms inference, 0.6ms postprocess per image at shape (1, 3, 480, 640)
[Edge 1] Detected: None (conf: 0.00), Accuracy: 0.97

0: 480x640 (no detections), 77.4ms
Speed: 1.3ms preprocess, 77.4ms inference, 0.5ms postprocess per image at shape (1, 3, 480, 640)
[Edge 1] Detected: None (conf: 0.00), Accuracy: 0.98

0: 480x640 (no detections), 78.8ms
Speed: 1.3ms preprocess, 78.8ms inference, 0.6ms postprocess per image at shape (1, 3, 480, 640)
[Edge 2] Detected: None (conf: 0.00), Accuracy: 0.99

0: 480x640 (no detections), 107.9ms
Speed: 1.5ms preprocess, 107.9ms inference, 0.8ms postprocess per image at shape (1, 3, 480, 640)
[Edge 2] Detected: None (conf: 0.00), Accuracy: 0.99

0: 480x640 (no detections), 129.5ms
Speed: 1.6ms preprocess, 129.5ms inference, 1.2ms postprocess per image at shape (1, 3, 480, 640)
[Edge 2] Detected: None (conf: 0.00), Accuracy: 0.98

```

(c) Cycle 5 - Edge 1 detection

```

0: 480x640 (no detections), 129.5ms
Speed: 1.4ms preprocess, 129.5ms inference, 1.2ms postprocess per image at shape (1, 3, 480, 640)
[Edge 2] Detected: None (conf: 0.00), Accuracy: 0.98

0: 480x640 (no detections), 81.4ms
Speed: 2.7ms preprocess, 81.4ms inference, 0.6ms postprocess per image at shape (1, 3, 480, 640)
[Edge 3] Detected: None (conf: 0.00), Accuracy: 0.99

0: 480x640 (no detections), 75.8ms
Speed: 1.4ms preprocess, 75.8ms inference, 0.7ms postprocess per image at shape (1, 3, 480, 640)
[Edge 3] Detected: None (conf: 0.00), Accuracy: 0.98

0: 480x640 (no detections), 75.2ms
Speed: 1.3ms preprocess, 75.2ms inference, 0.8ms postprocess per image at shape (1, 3, 480, 640)
[Edge 3] Detected: None (conf: 0.00), Accuracy: 0.98

[Cloud CI] Received 91 new frames for retraining.
[Cloud CI] Retraining complete + New Model v6 (Accuracy: 0.99)
[Edge 1] Model updated to v6, New Accuracy: 0.99
[Edge 2] Model updated to v6, New Accuracy: 0.99
[Edge 3] Model updated to v6, New Accuracy: 0.99

CI(YOLO) Simulation complete.
Process finished with exit code 0

```

(d) Cycle 5 - Edge 2 detection

Figure 3: Comparison of YOLO Edge Layer detections: Cycle 1 vs Cycle 5. The improvement in detection and accuracy is visible.

Interpretation

The results show that the Edge Layer YOLO-CI method can enhance its detection ability gradually through repeated rounds of data gathering, cloud-based retraining, and the deployment of the model. The system, even self-corrects if the first detections are few or wrong, thus, it reaches almost perfect accuracy at the later stages. Such a self-learning mode is an instance of the resilience of the integration of real-time perception at the edge with a CI-driven retraining loop in the cloud.

In addition, the Fog Layer's swarm-based coordination can rely on the improved accuracy at the edge to be a stable source of information, thus making the routing of emergency vehicles the result of the most accurate and up-to-the-minute detection data. In sum, the simulation is proof that the Edge-Cloud CI loop is the main driver behind the evolution of the static detection systems to become adaptive, intelligent perception modules that can be used for real-time, safety-critical traffic management tasks.

7.2 Part B: Fog Layer Swarm Intelligence

The simulation illustrates a hybrid Edge-Fog-Cloud traffic management system based on CNN-based vehicle detection, ACO-inspired swarm intelligence, and

centralized guidance for real-time ambulance routing. The simulation's main outcomes and insights are as follows:

1. Real-Time Traffic Updates at Edge Nodes

- Every intersection node updates its local traffic dynamically through a randomized model to model real-world traffic variances.
- Presence of emergency vehicles is randomly generated at each node at each timestep.
- Local nodes handle their own traffic and emergency status and exchange information with adjacent intersections to refresh pheromone trails.

Inference: Edge layer can autonomously perceive traffic and recognize emergencies while exchanging minimal information, lowering communication overhead.

2. Fog Layer Decision-Making using Swarm Intelligence

- Each fog node has pheromone levels for paths to nearby intersections.
- Pheromones are strengthened for routes with low traffic congestion or an emergency vehicle presence.
- Route choice employs a probabilistic method through pheromone density and traffic, emulating an Ant Colony Optimization (ACO) process.

Inference: The fog layer can autonomously make decisions such that there is real-time dynamic rerouting of emergency vehicles independent of the central server.

3. Central Server Global Guidance

- The central server is provided with cumulative traffic and emergency information from all the fog nodes.
- It calculates the world-wide average traffic and delivers pheromone updates to fog nodes to drive their local decision-making.

Inference: Global coordination thoughtfully directs local decisions to maximize global routing without congesting key routes.

4. Determination of Best Route

- The `find_best_route` method integrates pheromone intensity, traffic congestion, travel time, and emergency occurrence to probabilistically choose the optimal path.

- The system dynamically adjusts routes over time in accordance with real-time traffic changes and emergencies.

Example Output (Sample Timestep):

Timestep 5: Best Route=[1, 3, 4, 5], Cost=58.20, Avg Traffic=42.50

Inference: The system reliably finds a path of minimal overall travel cost and gives preference to emergency vehicle passage.

5. Pheromone Evolution

- Pheromone levels between intersections are revised at each timestep according to local traffic and emergency vehicle presence.
- Visualization illustrates how paths with high emergency vehicle passage or reduced congestion attain higher pheromone levels, strengthening optimal routes.

Inference: This illustrates adaptive learning, as the system reinforces improved paths dynamically and remembers poorer routes less in the long run.

6. Route Cost Analysis

- The graph of Best Route Cost over Time illustrates fluctuation that represents changes in traffic in real time.
- Decline in route cost indicates that the ACO-based pheromone reinforcement aids in converging into effective emergency paths across consecutive timesteps.

Inference: The system is resilient to dynamic traffic scenarios and responds effectively to evolving road conditions.

Key Observations

- Edge nodes support real-time, localized perception.
- Fog nodes support distributed decision-making using swarm intelligence.
- Central server provides global coordination without being a bottleneck.
- The system updates routes and pheromone trails dynamically to respond to emergencies effectively.
- Visualizations validate convergence to globally optimal routing while providing local autonomy.

```

Timestep 1: Best Route=[1, 3, 4, 5], Cost=186.00, Avg Traffic=53.60
Timestep 2: Best Route=[1, 2, 3, 4, 5], Cost=253.00, Avg Traffic=56.20
Timestep 3: Best Route=[1, 2, 3, 4, 5], Cost=230.00, Avg Traffic=52.80
Timestep 4: Best Route=[1, 2, 3, 4, 5], Cost=226.00, Avg Traffic=51.00
Timestep 5: Best Route=[1, 2, 3, 4, 5], Cost=221.00, Avg Traffic=51.40
Timestep 6: Best Route=[1, 3, 4, 2], Cost=163.00, Avg Traffic=52.40
Timestep 7: Best Route=[1, 3, 2, 4, 5], Cost=219.00, Avg Traffic=46.80
Timestep 8: Best Route=[1, 3, 4, 2], Cost=120.00, Avg Traffic=42.40
Timestep 9: Best Route=[1, 3, 4, 2], Cost=112.00, Avg Traffic=41.00
Timestep 10: Best Route=[1, 3, 4, 5], Cost=116.00, Avg Traffic=39.80

```

Figure 4: Output Snapshot

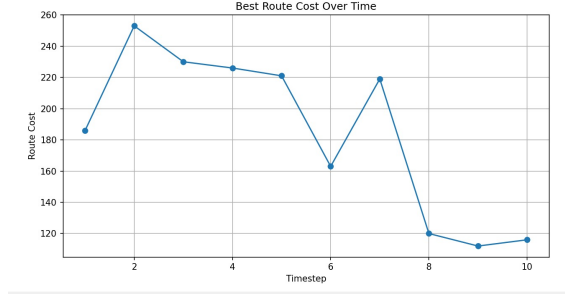


Figure 5: Output Graph

8 Conclusion

This report explains how computational intelligence (CI) can dramatically change real-time, mission-critical systems, using the example of the Intelligent Emergency Vehicle Prioritization System. The system, by deploying YOLOv8 at the Edge for on-the-spot detection and Swarm Intelligence (Ant Colony Optimization) at the Fog for coordinated traffic routing, represents a fully adaptive, CI-driven methodology that is capable of continuous learning, incorporating global feedback, and making decisions in real time.

The Edge layer utilizing YOLOv8 is forever changing the picture of the street with real-time video frames and detecting emergency vehicles in the most accurate way possible without the need for human assistance. This segment focuses on the computational intelligence perception facet, supplying tangible data to the Fog layer. At the Fog layer, swarm-based intersection agents bring in ACO to perpetually regulate traffic flow optimally. Each agent uses local traffic data, pheromone trails, and Edge detections to decide on the best routing that not only serves the immediate area but also takes into account the overall traffic situation. Although the Fog layer conducts its business in a distributed fashion of self-organization, it stays in tune with the global insights from the Cloud, thus enabling the best feasible coordination of the entire network of junctions.

The continuous intelligence loop is one of the primary features of this apparatus: Edge devices function as data collectors and processors, Fog agents perform routing optimization by utilizing both local and global feedback, whereas the Cloud acts as a data aggregator for retraining YOLO models and enhancing Swarm parameters. This cycle trains the system to keep getting better; it learns

new traffic scenarios and keeps on adapting so as to maintain the best possible emergency vehicle routing. Both YOLO and Swarm have been considered as two equally important computational intelligence units, thus exemplifying the dual and complementary roles of perception and coordination in real-time decision-making systems.

The system attains several significant results:

- Automated intelligence with minimal human intervention: Emergency vehicle detection and traffic signal coordination activities are carried out smoothly, thereby shortening response times, and operations can go on without interruption even in highly dynamic traffic scenarios.
- Adaptive, self-improving coordination: Swarm agents never stop tweaking their routing methods according to updated pheromone trails and worldwide feedback, thus ensuring that traffic is flowing smoothly and that emergency vehicles are given dynamic priority.
- Continuous enhancement and model precision: The Edge CI loop guarantees that YOLO models are regularly retrained using the Cloud data, thus the detection capability is always kept at a high level and is strong enough to handle different situations.
- Operations that are distributed yet in sync: The different layers, i.e., Edge, Fog, and Cloud, are working in harmony to attain not only local responsiveness but also global optimization, thus providing a scalable approach to urban traffic intelligence.

The work implemented here is a case in point of how computational intelligence is far from being just a theoretical idea but rather a key that opens the door to practical, real-world urban mobility solutions. The proceeding from the Edge perception intelligence to the Fog-based coordination intelligence is an integrated working model that is robust, adaptable, and scalable, and it solves the intricate, volatile problems of emergency vehicle prioritization.

The system is packed with a lot of potential for future expansion and upgrades. Some of the potential future enhancements entail a YOLO model federated learning under which various Edge nodes can jointly carry out model updates without sharing raw data, thus ensuring data privacy and lessening the network load. The Swarm coordination may be widened to multi-agent, city-scale networks to optimize the whole urban area better. Besides that, trend analysis of traffic could be employed to forecast congestion, which in turn would further the road emergency response. The eagerness to know what is happening in the surrounding environment may be satisfied by adding more Edge sensors, such

as LiDAR or GPS tracking, while reinforcement learning techniques may be used to perfect Swarm behavior for more intricate traffic scenarios. In conclusion, this research serves as an example of how a computational intelligence-based system that integrates YOLO and Swarm in an Edge-Fog-Cloud architecture can be used for traffic management that is real-time, adaptive, and efficient for emergency vehicles. It proves that combining perception and coordination intelligence is the best way to arrive at a self-improving, intelligent system that not only drastically cuts the emergency response time but also provides the basis for the upcoming smart city traffic solutions. This endeavor offers a scalable blueprint of the possible ways by which computational intelligence can be brought to bear in safety-critical environments of the real world, thus demonstrating that the advanced artificial intelligence and optimization algorithms can have tangible, life-saving impacts on urban infrastructure. In essence, the system showcases the real-world power of computational intelligence to enable faster, safer, and more efficient emergency response in smart cities.

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